

Shomi Ahmed, MSc, EIT

Electrical Engineer | Hardware Systems | Mixed-Signal, Embedded & PCB Design

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Ottawa, ON | Canadian Citizen | Eligible for Reliability & Secret Clearance

PROFESSIONAL SUMMARY

Electrical Engineer with 5+ years of experience designing precision analog, embedded, and multi-layer PCB hardware for detector instrumentation and data acquisition systems. Developed ultra-stable current sources (<1 ppm, ~1 nA drift) and 64-channel control systems using SPI-based architectures and precision DACs. Experienced across the full hardware lifecycle from requirements and PCB design to validation and deployment, with expertise in mixed-signal design, high-speed PCB layout, and reliable operation in high-voltage, low-noise environments, with a strong focus on testing and system reliability.

TECHNICAL SKILLS

Hardware Design: Multi-layer PCB design (up to 8 layers), mixed-signal systems, controlled impedance routing, differential signaling, signal integrity, grounding & shielding, EMI-aware layout

Analog & Power: Precision current sources (<1 ppm stability), feedback control systems, loop stability, low-noise design (OPA227, INA828, LTC6655), MOSFET-based power control

Embedded Systems: Microcontrollers (Teensy, Arduino, Raspberry Pi Pico), SBC platforms (CM5, NanoPi), embedded control and system integration

Interfaces & Protocols: SPI, I²C, UART, multi-channel DAC systems (LTC2668), ASIC/FPGA interfacing (IpGBT)

Verification & Testing: Oscilloscope, spectrum analyzer, waveform generator, test procedures, validation, performance characterization, root-cause analysis

Tools & Simulation: KiCad, Altium Designer, LTSpice, SolidWorks, Python, C/C++

Manufacturing & Documentation: DFM/DFT, stack-up definition, Gerber generation, supplier interfacing, PCB assembly, requirements definition, electrical interface documentation (EICD)

PROFESSIONAL EXPERIENCE

Electrical Engineer – Carleton University, Ottawa, ON

Oct 2025 – Present

- Lead development of electronics for detector instrumentation and DAQ systems for nEXO and Hyper-K, including NanoPi-based systems for SiPM biasing and amplification
- Perform system-level integration across sensors, readout electronics, embedded controllers, and SBC platforms
- Design SiPM carrier boards for signal acquisition and routing, including interface definition and signal mapping
- Develop embedded control architecture (Teensy + CM5) replacing hardware extenders to enable reliable remote camera operation and system-level control
- Engineer hardware interface for Sony A7R IV electronic control via SSR-based power switching
- Support mPMT detector testing, including installation and integration of main electronics boards
- Perform hardware bring-up, debugging, verification, and root-cause analysis
- Support procurement, assembly, and deployment of experimental hardware
- Collaborate with mechanical, firmware, and multidisciplinary teams for system integration

Electronics Specialist – University of Winnipeg, MB

Nov 2020 – Oct 2025

- Delivered end-to-end hardware systems for large-scale instrumentation projects (TUCAN, Hyper-K, MOLLER, DEAP, nEXO), from requirements definition through deployment
- Led full development lifecycle including schematic capture, PCB design, fabrication, assembly, validation, and field deployment, including on-site commissioning at TRIUMF and CERN
- Designed ultra-stable ± 200 mA precision current source (<1 ppm stability, ~1 nA drift) using OPA227, INA828, LTC6655, and MOSFET-based feedback, exceeding commercial instrument performance
- Performed LTSpice simulation, loop stability analysis, and precision component selection for high-performance analog systems (LTC6655 reference, low-TCR resistors)
- Architected and implemented 64-channel control system (± 40 mA/channel) using LTC2668 DACs with SPI-based control, achieving 10^{-5} system stability

- Developed multi-board system architecture (power, control, and current boards) with defined electrical interfaces for scalable deployment
- Designed and implemented multi-layer PCBs (up to 8 layers) for mixed-signal and high-speed systems, applying signal integrity, grounding, and manufacturability best practices
- Implemented communication interfaces (SPI, I²C, UART) and contributed to ASIC/FPGA interfacing for IpGBT-based readout systems
- Interfaced with PCB manufacturers to refine stack-up, DFM constraints, and fabrication outputs
- Led hardware bring-up, debugging, verification, and validation, including development of test setups, procedures, and performance characterization
- Delivered 45+ PCBs, including assembly (QFN), verification, and deployment
- Investigated hardware anomalies and conducted root-cause analysis to improve system reliability
- Designed electronics for high-voltage systems (± 250 kV) with pA-level sensitivity, ensuring stable operation in low-noise environments
- Mentored 10+ students and contributed to lab infrastructure development, equipment procurement, and supplier coordination

KEY PROJECTS

64-Channel Ultra-Stable Current Supply

- Designed and implemented ± 40 mA/channel (64-channel) current control system using LTC2668 DACs
- Developed SPI-based control architecture (Arduino) for synchronized multi-channel operation
- Achieved 10^{-5} stability for precision magnetic field control applications
- Defined multi-board architecture (power, control, and current boards) with integrated signal interfaces
- Full paper accepted for oral presentation at IEEE CCECE 2026

Ultra-Stable Precision Current Source

- Designed ± 200 mA precision current source with <1 ppm stability (~ 1 nA drift)
- Implemented analog feedback topology using OPA227, INA828, LTC6655, and MOSFET-based current control
- Performed LTSpice simulation, loop stability analysis, and experimental validation
- Achieved stability and noise performance exceeding commercial instruments
- Published and presented at [IEEE MELECON 2024](#)

8-Layer High-Speed Readout Board (IpGBT-based)

- Designed and implemented 8-layer mixed-signal PCB with controlled impedance and differential signaling
- Integrated IpGBT-based ASIC/FPGA interfaces for high-speed data acquisition
- Applied stack-up and layout constraints to ensure signal integrity and manufacturability

IWCD Photogrammetry Camera Control System (CM5 + Teensy)

- Designed embedded control architecture using CM5 and Teensy, replacing hardware extenders
- Integrated SBC and microcontroller platforms for automated control and remote operation
- Interfaced with internal camera power circuitry for hardware-level control via SSR-based switching
- Enabled reliable operation of Sony A7R IV imaging system in constrained experimental environments

EDUCATION

M.Sc. (Physics) – University of Manitoba, Winnipeg, MB

Jan 2016 – Oct 2019

- CGPA: 4.0 / 4.5; Manitoba Graduate Scholarship (2017–2018, Rank: 2nd)
- Focus: Precision instrumentation and measurement systems

B.Sc. (Electrical & Electronic Engineering) – Islamic University of Technology, Bangladesh

Jan 2010 – Oct 2013

- CGPA: 3.91 / 4.0

PROFESSIONAL AFFILIATIONS & SERVICE

- Engineer-in-Training (EIT), Engineers Geoscientists Manitoba (#46997) Nov 2021 – Present
 - Publication Chair, IEEE Ottawa Section Feb 2026 – Present
 - Former Student Activities Committee Chair, IEEE Winnipeg Section Feb 2025 – Jan 2026
 - IEEE Member – Session Chair and Technical Reviewer (International Conferences)
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